

SQLite and PostgreSQL basics

Module 4 - Lesson 3

Overview

Databases store data durably. SQLite is a file-based database for small apps; PostgreSQL is a full server for production. Both speak SQL and integrate with Python.

Key Concepts

- SQLite is embedded in Python via the sqlite3 module; no server required.
- Create tables with CREATE TABLE and insert with INSERT INTO ... VALUES.
- Parameterized queries (? placeholders) prevent SQL injection.
- SELECT, WHERE, ORDER BY, and JOIN query data with precision.
- PostgreSQL adds concurrency, advanced types, and is used via a driver like psycopg.
- ORMs (SQLAlchemy, Drizzle, Prisma) map tables to objects and manage migrations.

Example

```
import sqlite3

conn = sqlite3.connect("app.db")
conn.execute("CREATE TABLE IF NOT EXISTS courses (id INTEGER PRIMARY KEY, title TEXT)")
conn.execute("INSERT INTO courses (title) VALUES (?)", ("Python",))
conn.commit()

for row in conn.execute("SELECT * FROM courses ORDER BY id"):
    print(row)
```

Summary

SQLite stores data in a single file; PostgreSQL scales to production. Parameterized queries keep data safe, and ORMs make schemas and queries maintainable.

Practice Checklist

- Create a SQLite table and insert three rows with parameters.
- Query rows with a WHERE filter and ORDER BY.
- Compare when you would choose SQLite versus PostgreSQL.